Projection Pursuit and Lasso Regression: A Comparison

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Abstract. Dimension reduction is used to perform regression on large dimension datasets. We define the practice of projection pursuit regression. We then perform a simulation study to compare the predictive ability of lasso and projection pursuit regression models. Finally, we apply projection pursuit and lasso regression methods to a dataset of Indian demographics to compare the strengths and weaknesses of each type of analysis

Keywords: multiple regression, dimension reduction, projection pursuit, LASSO

1 Introduction

Dimension reduction for use in regression has gained importance as faster computers give us access to large data sets. As the number of variables we can gather data on increases, variable selection by the standard method of iterative selection becomes increasingly impossible.

In this paper, we explore the strengths and weaknesses of using projection pursuit regression as a dimension reduction technique. Currently, projection pursuit regression is largely ignored except for its connection to neural networks and as a fringe case for nonparametric regression. Also, projection pursuit regression is less popular because it involves high computational demands. As processing speeds have increased, however, projection pursuit regression presents a possible alternative to current dimension reduction techniques with regards to predictive models.

Our examination of projection pursuit regression as a dimension reduction technique is two-pronged. First, we examine the relative efficiency of projection pursuit regression to handle data where the number of observations n is less than the dimension p under various circumstances. Second, we use projection pursuit regression and lasso regression to analyze a real data set of demographics about political districts in India where $n < p$, trying to gauge the model's ability to predict child mortality rate.

The simulation study will evaluate two different data scenarios at different sample size, correlation, variance, and dimension levels. The first data scenario will evaluate how each dimension reduction technique acts when only certain covariates are significant with linear slopes in a data set. The other will evaluate dimension reduction techniques in reference to when a relationship exists between all covariates and the response variable, along with nonlinear effects being present in the data. The first scenario should favor lasso and ridge regression techniques, while the second will presumably favor projection pursuit regression.

The application study will then be performed using lasso and projection pursuit regression, with a discussion on the strengths and weaknesses of either approach. The goal of the analysis is to predict child mortality rates within a Indian district.

2 Methods

2.1 Projection Pursuit Regression

Projection pursuit regression is viable for dimension reduction because it reduces the number of parameters that need to be estimated. Instead of finding slope coefficients for all of the covariates included in the model, projection pursuit seeks M unit vectors that are descriptive of trends in the data. Once these trends have been estimated, a nonlinear function S_m is applied to the product of the unit vector $\boldsymbol{\omega}_m$ and our matrix of covariates $\boldsymbol{X}$, such that Equation 1 is minimized.

$$\sum_{i=1}^{N} \left[y_i - \sum_{m=1}^{M} S_m(\boldsymbol{\omega}_m^T x_i) \right]^2. \tag{1}$$

To do this, we assume the regression surface is the sum of univariate functions on the predictors.

$$\varphi(\boldsymbol{X}) = \sum_{m=1}^{M} S_m(\boldsymbol{\omega}_m \cdot \boldsymbol{X}). \tag{2}$$

Also we center the response variable y_i at 0. Projection pursuit regression consists of then searching for a projection $\boldsymbol{\omega}_{M+1}$ that maximizes the *figure of merit* $I(\boldsymbol{\omega})$.

$$I(\boldsymbol{\omega}) = 1 - \sum_{i=1}^{n} (r_i - S_{\boldsymbol{\omega}}(\boldsymbol{\omega} \cdot \boldsymbol{X}))^2 / r_i^2. \tag{3}$$

For the first iteration, $r_i = y_i$ and $M = 0$. We find $S_{\boldsymbol{\omega}}$ such that, given $\boldsymbol{\omega} \cdot \boldsymbol{X}$, the residuals are minimized. We then find a $\boldsymbol{\omega}_{M+1}$ to minimize the figure of merit $I(\boldsymbol{\omega}_{M+1})$. If, however, after the first iteration the figure of merit $I(\boldsymbol{\omega}_{M+1})$ is still above a predetermined threshold, redefine $r_i = r_i - S_{\boldsymbol{\omega}_{M+1}}(\boldsymbol{\omega}_{M+1} \cdot \boldsymbol{X})$ and $M = M+1$ to begin the process again until the figure of merit is acceptable. The last term is not included.

Note that projection pursuit regression assumes no distribution of the data $\boldsymbol{y}$. Projection pursuit regression was developed primarily for nonparametric data analysis, similar to the k-means algorithm and other nonparametric approaches to data analysis. In this analysis we seek to evaluate how projection pursuit regression performs when the true distribution is the multivariate normal, and whether projection pursuit regression is a viable option relative to other dimension reduction methods usually applied to normally distributed random variables.

2.2 Ridge & Lasso Regression

Traditional methods of dimension reduction include ridge and lasso regression. Specifically, we seek to shrink the number of covariates we will use in our analysis. Ridge and lasso regression are more elegant than stepwise variable selection and allow the user to fit models even when there are far more covariates than there are total observations.

We consider ridge regression first of the two dimension reduction methods. Specifically, we seek the solution to a penalized least squares equation $\hat{\beta}^{\text{ridge}}$.

$$\hat{\beta}^{\text{ridge}} = \text{argmin}_{\beta} \left\{ \sum_{i=1}^{N} (y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}. \tag{4}$$

The parameter λ is decided using cross-validation results. Notice, that we assume no distribution for this kind of variable selection, so we have no assumptions to fulfill in order to use this tool. Ordinarily the covariates are centered and scaled, which we will do in our simulation study.

Lasso regression is similar to ridge regression, but is distinct enough from ridge regression to gain some advantages. Lasso regression finds the solution to a penalized least squares equation $\hat{\beta}^{\text{lasso}}$.

$$\hat{\beta}^{\text{lasso}} = \text{argmin}_{\beta} \sum \left(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j \right)^2 \text{ subject to } \sum_{j=1}^{p} |\beta_j| \leq s. \tag{5}$$

We calibrate s using cross validation, maximizing model fit and predictive ability. Lasso is preferable to ridge regression because the coefficients it determines are not significant are zeroed out naturally, removing them from the regression equation.

3 Simulation Study

In performing dimension reduction we are primarily concerned with problems of model fit. Too many components in an analysis allow for an overfit model with a raised likelihood but poor predictive ability. Overfit models are often inaccurate about the size of effects when performing inference. Root predicted mean squared error (RPMSE) is a good indicator of overfit. If the model fits and predicts well, RPMSE will be low and indicate a good fit. Also, because projection pursuit regression has no assumed distribution, we cannot find the

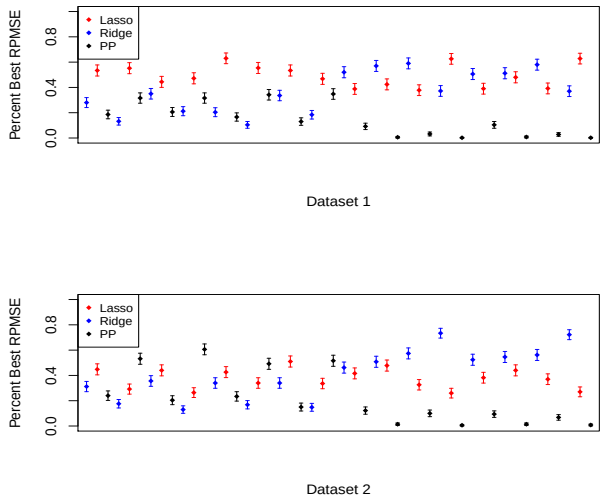

Fig. 1. Proportion Each Regression Approach Performed Optimally by RPMSE with Monte Carlo Intervals

	n	ρ_1	ρ_2	σ^2	p
1	20	0.0	0.1	0.5	20
2	50	0.0	0.1	0.5	20
3	20	0.5	0.9	0.5	20
4	50	0.5	0.9	0.5	20
5	20	0.0	0.1	2	20
6	50	0.0	0.1	2	20
7	20	0.5	0.9	2	20
8	50	0.5	0.9	2	20
9	20	0.0	0.1	0.5	200
10	50	0.0	0.1	0.5	200
11	20	0.5	0.9	0.5	200
12	50	0.5	0.9	0.5	200
13	20	0.0	0.1	2	200
14	50	0.0	0.1	2	200
15	20	0.5	0.9	2	200
16	50	0.5	0.9	2	200

Table 1. Reference Table for Simulation Study with Alternate ρ Values for Each Data Set

likelihood in order to use common diagnostic values like the Akaike Information Criterion and the Bayesian Information Criterion.

We therefore compare the RPMSE in a cross validation study for each of our regression approaches. The lowest RPMSE value will indicate the optimal fit out of the models we consider. We are primarily concerned with the relative performance of lasso regression, ridge regression, and projection pursuit regression. We simulate datasets and then perform a cross validation study, training the each model using 90% of the simulated data in order to predict the response variable for the remaining tithe of the dataset. Each approach uses the same 90% of the dataset to train the model. We then compare the RPMSE values for each approach and report the most optimal approach.

For all our fabricated data, we keep the values for both $\boldsymbol{X}$ and $\boldsymbol{y}$ centered. All three algorithms work optimally when the data are centered around 0. We drew covariates $\boldsymbol{X}$ from a multivariate normal distribution with mean $\mathbf{0}$ and covariance matrix $\boldsymbol{\Sigma}$. $\boldsymbol{\Sigma}$ had an Autoregressive[1] structure with parameter ρ, such that, for correlation structure $\boldsymbol{H}$ with parameter ρ, $\boldsymbol{\Sigma} = \sigma^2\boldsymbol{H}$. $\boldsymbol{y}$ was then calculated by applying a vector of slopes $\boldsymbol{\beta}$ to our simulated data and adding $\boldsymbol{z} \sim N(0,1)$, to incorporate some randomness into our response variable.

Table 2. RPMSE Means - Data 1

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Lasso	1.195	1.070	1.151	1.038	1.277	1.046	1.118	1.030	3.030	3.000	3.443	2.036	5.843	5.719	6.661	3.300
Ridge	1.362	1.206	1.178	1.085	1.631	1.237	1.230	1.104	2.887	2.873	2.936	2.232	5.608	5.581	5.515	3.857
PP	1.867	1.219	1.872	1.188	2.157	1.183	2.249	1.150	6.716	7.922	10.039	11.596	12.380	15.260	20.288	22.971

Table 3. RPMSE Means - Data 2

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Lasso	4.817	4.552	5.202	4.614	30.535	27.215	30.568	28.470	16.294	16.959	23.354	23.394	110.000	110.249	135.441	144.432
Ridge	4.890	4.467	5.206	4.766	30.399	27.565	31.249	29.559	15.597	16.777	20.623	19.559	105.935	106.067	124.041	124.614
PP	6.377	4.249	7.289	4.308	40.882	26.546	44.385	28.251	32.989	41.048	49.848	60.189	225.764	276.710	307.415	377.439

We change $\boldsymbol{\beta}$ to create two different types of datasets for each level of n, p, ρ, and σ^2. We do so to compare the regression approaches, creating a base simulation for each regression approach; one set of simulations that favor lasso regression, and another set of simulations that should favor projection pursuit regression.

In order to give lasso regression an advantage, we produce $\boldsymbol{y}$ using the equation $\boldsymbol{y} = \boldsymbol{X\beta} + \boldsymbol{z}$. $\boldsymbol{\beta}$ is a vector of 0's and 1's, where $p/10$ elements of $\boldsymbol{\beta}$ are 1. Lasso regression will presumably be able to zero out the covariates where $\boldsymbol{\beta} = 0$. Projection pursuit regression should be less advantageous with this set of simulated data.

To produce a group of datasets where projection pursuit regression is optimal, we produce $\boldsymbol{y}$ using the equation $\boldsymbol{y} = \boldsymbol{X}^*\boldsymbol{\beta}^* + \boldsymbol{z}$. $\boldsymbol{X}^*$ includes $2p/5$ extra columns, incorporating the square and cube of each of $p/5$ randomly selected columns. The inclusion of nonlinear elements should make projection pursuit regression more advantageous.

As previously mentioned, we produce two groups of datasets, not just two datasets. We do this to see the effect different levels of correlation ρ, observation count n, and dimension p have on RPMSE for each regression approach. The simulation process is computationally intensive, with 500 iterations demanding 12 hours on three computer cores. Because of this, our simulation is limited to 500 iterations.

Note that the vector $\boldsymbol{y}$ will be centered around 0, even if the overall scale is different from dataset to dataset. We do this to ensure that lasso regression and projection pursuit regression, which perform optimally when the data is centered, face no unforeseen problems in fitting to the data.

Figure 1 shows the results of the simulation study. Trials are referenced by Table 1. For our first data set, where lasso and ridge regression should be optimal, we see that when n was large and p was relatively small projection pursuit performed relatively well. However, when p was increased to far greater than n, projection pursuit regression practically never performed optimally.

For our second data set, where projection pursuit was supposed to be optimal, the results are disconcerting. As can be seen, projection pursuit and lasso regression performed optimally a similar percentage of the time when $n \geq p$. However, despite our attempt to create data advantageous to projection pursuit regression, lasso and ridge regression vastly outperformed projection pursuit regression with regards to predictive ability when $n < p$.

From our simulation, we can conclude that lasso regression and ridge regression are optimal for dimension reduction on normally distributed data with regards to predictive ability. Changes in correlation parameter ρ and variance parameter σ^2 did not appear to make a large difference between which regression approach was optimal.

4 Application

We explore the data before beginning any applications. There are 284 districts of India in this data set, each with 403 demographic measures attached. For the purposes of this analysis, we removed the rows and columns that had missing data. Further, we removed all columns corresponding to variables that were summaries of

	$\boldsymbol{\omega}_1$		$\boldsymbol{\omega}_1$
CBR R	0.18	Safe Delivery U	0.16
Baby Food Fruit/Veg R	0.15	Pills Usage U	0.15
Children No Unvaccinated U	0.15	Sample Units R	-0.15
Chronic Ill Treat Female R	-0.14	Sex Ratio U	0.13
Home Birth Mid U	-0.13	Sex Ratio R	-0.12
1 Month Food Veg/Fruit R	-0.12	Delivery in Gov Hosp U	-0.12
Nat Grow Rate R	-0.12	Mothers Hosp R	0.11
Mothers in Non-gov Hosp U	0.11	Mothers Blood Taken U	-0.11
Ultrasound R	0.11	Women Zinc R	-0.11
Young Birth Order 2 R	0.11	Arthritis Female U	-0.10
Treat Any Source Female R	0.10	Avg Month Abort U	-0.10
Mothers Antenatal U	0.10		

Table 4. Selected Coefficients from ω_1

	Estimate	LB	UB
Crude Birth Rate R	2.22	1.42	3.01
Intercept	-281.74	-383.35	-180.13
Baby Food Fruit/Veg R	1.01	0.47	1.56
Children Unvaccinated R	1.35	0.60	2.10
Children Resp Infection R	0.29	0.10	0.47
Median FLB R	4.99	1.53	8.46
TB Infection Male R	0.02	0.01	0.04
Delivery in Gov Hosp U	0.27	0.06	0.48
PostNatal Care 1 Week R	0.32	0.05	0.59
Sex Ratio U	0.03	0.00	0.05
No Financial Aid R	0.22	0.01	0.43
Asthma Female R	0.01	0.00	0.01

Table 5. Coefficients for Lasso Regression on Indian Districts

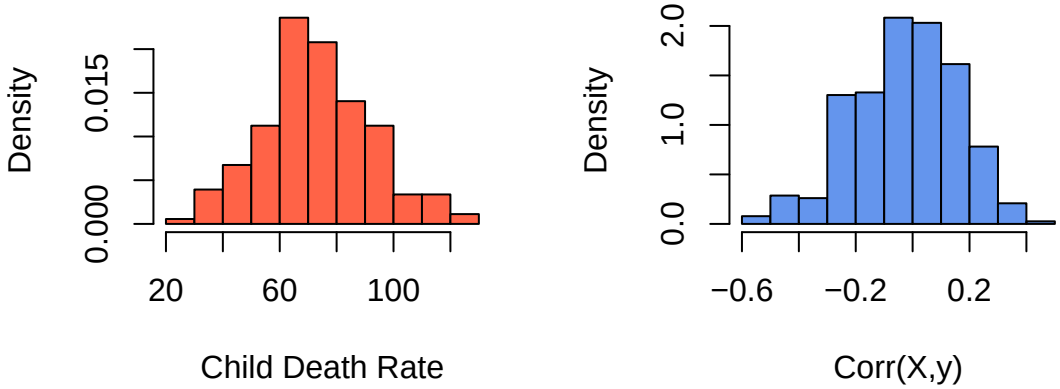

Fig. 2. Exploratory Data Plots

other columns, and columns that were components of child mortality rates. The resulting data set had 384 columns for 178 districts of India.

We see that many columns of the data are not highly correlated with our response variable, so we can avoid problems with collinearity. (Figure 2) Also, the response variable seems to be normally distributed. This data therefore seems perfectly suited to compare lasso and projection pursuit regression on normal data. In order to ensure the variables were conducive to lasso and projection pursuit regression, we standardize the covariates and the response variable.

We first used lasso regression on the data set. Regression coefficients that were significantly different than 0 are shown in Table 3. 95% confidence bounds are also included. Coefficient names are not extremely descriptive because of constraints of space. However, many of the coefficients included are linked with disease and hygiene.

Projection pursuit regression found that one projection produced the best model, such that there was only one $\boldsymbol{\omega}_1$ unit vector which $\boldsymbol{X}$ was projected onto. The largest elements of $\boldsymbol{\omega}_1$ are found in Table 4. Note that a spline is then applied to $\boldsymbol{\omega}_1 \cdot \boldsymbol{X}$. Because of this transformation, $\boldsymbol{\omega}_1$ is not interpretable in a useable way. However, it is interesting to note that a fair few of the significant coefficients in the lasso regression also have a large presence in the projection $\boldsymbol{\omega}_1$. That is, a spline on a function of effects that appear to be significant produces the best projection pursuit regression model.

The lasso regression model predicts better than the projection pursuit regression model. A cross-validation study was repeated 100 times by predicting 10% of the data with the remaining 90% of the data. The average RPMSE value was 0.637. A similar study was conducted for the projection pursuit regression model with an average RPMSE of 0.962. We see that both models fit rather well, but the lasso regression model is better suited to making predictions of other districts in India.

Interestingly, the projection pursuit model fits better to the data within the model. The Root Mean Square Error (RMSE) for the projection pursuit regression model is 7.26×10^{-17}, significantly less than the

Table 6. Cross-Validation RPMSE and Bootstrap RMSE Results

	Full Data RMSE	Average	95% LB	95% UB
Lasso RMSE	0.555	0.476	0.393	0.587
Projection Pursuit RMSE	9.98×10^{-17}	1.04×10^{-14}	2.58×10^{-15}	1.38×10^{-7}
Lasso RPMSE	-	0.678	0.449	0.937
Projection Pursuit RPMSE	-	0.984	0.713	1.241

lasso regression model's RMSE of 0.56. The diagnostic measures of the projection pursuit regression model relative to the lasso regression model suggest that the model is overfit even though $M = 1$.

We now confront the elephant in the room, that we have not properly defined a research question to answer in performing analysis on this dataset. If the goal is prediction, it appears that projection pursuit regression and lasso regression are equally advantageous. However, because these data are distributed normally, a lasso regression allows for performing inference. Projection pursuit regression is a predictive machine learning tool, so no inference is possible.

For this analysis, we gain nothing by reducing the dimension by projection pursuit regression rather than lasso regression. As shown in our simulation study, projection pursuit regression doesn't predict better than lasso regression. Also, we lose all ability to perform inference.

5 Conclusions

We see that projection pursuit regression is inappropriate for use as a dimension reduction tool in multiple linear regression on normally distributed data. While projection pursuit regression is not viable as the optimal tool for dimension reduction, we see from our simulation study that when dimension reduction is not necessarily needed ($n < p$) projection pursuit regression predicts rather well.

As a predictive tool, projection pursuit regression appears a viable tool when dimension reduction is not needed. As we saw in analyzing our real data set, projection pursuit predicts rather well and fits the data extremely well. However, like other machine learning techniques, we cannot make any inference on the effects of covariates on the response variable. Because of this and the lack of predictive accuracy when p is large, projection pursuit regression is best applied to nonparametric data sets where a distribution of any kind is not justifiable.

While we found that projection pursuit regression is comparable to lasso and ridge regression processes when $n > p$, the inability to perform inference suggests that, generally, analyzing normal data using projection pursuit regression is a mistake. Lasso and ridge regression are more optimal for normally distributed data.

Future steps to explore the application of projection pursuit regression would include simulation studies on data where $n > p$. Projection pursuit regression could perform better on data distributed according to an exponential family distribution with nonlinear elements in a regression sense. For example, a comparison of predictive ability between a Poisson regression model and a projection pursuit regression model would be interesting. Furthermore, a comparison when interaction terms are prolific throughout a dataset. Friedman and Stuetzle[3] originally used a dataset with interaction terms to motivate projection pursuit regression's introduction. It would be interesting to see a simulation study similar to the one performed above that incorporates interaction terms instead of nonlinearity.

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